AdapLeR: Speeding up Inference by Adaptive Length Reduction

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Abstract

Pre-trained language models have shown stellar performance in various downstream tasks. But, this usually comes at the cost of high latency and computation, hindering their usage in resource-limited settings. In this work, we propose a novel approach for reducing the computational cost of BERT with minimal loss in downstream performance. Our method dynamically eliminates less contributing tokens through layers, resulting in shorter lengths and consequently lower computational cost. To determine the importance of each token representation, we train a Contribution Predictor for each layer using a gradient-based saliency method. Our experiments on several diverse classification tasks show speedups up to 22x during inference time without much sacrifice in performance. We also validate the quality of the selected tokens in our method using human annotations in the ERASER benchmark. In comparison to other widely used strategies for selecting important tokens, such as *saliency* and *attention*, our proposed method has a significantly lower false positive rate in generating rationales. Our code is freely available at https://github.com/amodaresi/AdapLeR.

1 Introduction

While large-scale pre-trained language models exhibit remarkable performances on various NLP benchmarks, their excessive computational costs and high inference latency have limited their usage in resource-limited settings. In this regard, there have been various attempts at improving the efficiency of BERT-based models (Devlin et al., 2019), including knowledge distilation (Hinton et al., 2015; Sanh et al., 2019; Sun et al., 2019, 2020; Jiao et al., 2020), quantization (Gong et al., 2014; Shen et al., 2020; Tambe et al., 2021), weight pruning (Han et al., 2016; He et al., 2017; Michel et al., 2019; Sanh et al., 2020), and progressive module replacing (Xu et al., 2020). Despite providing significant reduction in model size, these techniques are generally static at inference time, i.e., they dedicate the same amount of computation to all inputs, irrespective of their difficulty.

A number of techniques have been also proposed in order to make efficiency enhancement sensitive to inputs. *Early exit* mechanism (Schwartz et al., 2020b; Liao et al., 2021; Xin et al., 2020; Liu et al., 2020; Xin et al., 2021; Sun et al., 2021; Eyzaguirre et al., 2021) is a commonly used method in which each layer in the model is coupled with an intermediate classifier to predict the target label. At inference, a halting condition is used to determine whether the model allows an example to exit without passing through all layers. Various halting conditions have been proposed, including Shannon's entropy (Xin et al., 2020; Liu et al., 2020), softmax outputs with temperature calibration (Schwartz et al., 2020b), trained confidence predictors (Xin et al., 2021), or the number of agreements between predictions of intermediate classifiers (Zhou et al., 2020).

Most of these input-adaptive techniques compress the model from the depth perspective (i.e., reducing the number of involved encoder layers). However, one can view compression from the width perspective (Goyal et al., 2020; Ye et al., 2021), i.e., reducing the length of hidden states. (Ethayarajh, 2019; Klafka and Ettinger, 2020). This is particularly promising as recent analytical studies showed that there are redundant encoded information in token representations (Klafka and Ettinger, 2020; Ethayarajh, 2019). Among these redundancies, some tokens carry more task-specific information than others (Mohebbi et al., 2021), suggesting that only these tokens could be considered through the model. Moreover, in contrast to layer-wise pruning, token-level pruning does not

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